

Parking Management System

BCA Mini-Project Documentation

1. Step-by-Step Algorithm

1. Start.
2. Enter vehicle details such as Vehicle Number and Vehicle Type.
3. Check available parking slots.
4. If a slot is available, assign a parking slot.
5. Record the vehicle entry date and time.
6. Store the vehicle and parking details.
7. Keep the vehicle in the assigned parking slot.
8. When the vehicle exits, enter the Vehicle Number.
9. Retrieve the vehicle entry time.
10. Record the vehicle exit date and time.
11. Calculate the total parking duration.
12. Update the parking slot status as Available.
13. Store the completed parking details in Parking History.
14. Display the parking information and parking duration.
15. Stop.

2. Prompt for Flowchart

Create a professional, clean, well-organized flowchart for a "Parking Management System".

The flowchart should represent the complete process from vehicle entry to vehicle exit and parking history.

Use standard flowchart symbols:

Oval = Start/Stop

Parallelogram = Input/Output

Rectangle = Process

Diamond = Decision

Arrows = Flow direction

Flowchart steps:

START ↓ Enter Vehicle Details

(Vehicle Number, Vehicle Type) ↓ Check Available Parking Slot ↓ Is Parking Slot Available? → NO:

Display "No Parking Slot Available" → STOP → YES: Assign Parking Slot ↓ Record Entry Date and Time ↓

Store Vehicle and Parking Details ↓ Vehicle Remains Parked ↓ Vehicle Exit Request ↓ Enter Vehicle

Number ↓ Retrieve Entry Time ↓ Record Exit Date and Time ↓ Calculate Parking Duration ↓ Update Parking

Slot as Available ↓ Save Details in Parking History ↓ Display Parking Details and Duration ↓ STOP

Make the diagram vertical from top to bottom, use clear arrows, properly aligned boxes, readable text, standard symbols, and good spacing. Give it an academic mini-project style suitable for a BCA college project.

3. Prompt for ER Diagram

Create a professional ER (Entity Relationship) diagram for a “Parking Management System” based on these features: Vehicle Details, Parking Slot, Entry/Exit, Parking Duration, and Parking History.

Use Chen ER diagram notation with rectangles for entities, ovals for attributes, diamonds for relationships, and clearly marked primary keys.

Create these main entities:

1. VEHICLE
Vehicle_Number (Primary Key)
Vehicle_Type
Owner_Name

2. PARKING_SLOT
Slot_ID (Primary Key)
Slot_Number
Slot_Status

3. PARKING_RECORD
Record_ID (Primary Key)
Vehicle_Number (Foreign Key)
Slot_ID (Foreign Key)
Entry_Date
Entry_Time
Exit_Date
Exit_Time
Parking_Duration

4. PARKING_HISTORY
History_ID (Primary Key)
Record_ID (Foreign Key)
Vehicle_Number
Slot_ID
Entry_Time
Exit_Time
Parking_Duration

Show these relationships clearly:
VEHICLE is assigned to PARKING_SLOT
VEHICLE has PARKING_RECORD
PARKING_SLOT is used in PARKING_RECORD
PARKING_RECORD is stored in PARKING_HISTORY

Show appropriate cardinalities such as 1:N where applicable.
Arrange the entities neatly with no overlapping lines. Clearly label Primary Key (PK) and Foreign Key (FK). Use a clean white background, professional academic style, readable fonts, balanced spacing, and high-resolution output suitable for a BCA mini-project documentation.

Simple ER Structure

VEHICLE PK Vehicle_No Vehicle_Type Owner_Name	PARKING_RECORD PK Record_ID FK Vehicle_No FK Slot_ID Entry_Date Entry_Time Exit_Date Exit_Time Parking_Duration	PARKING_HISTORY PK History_ID FK Record_ID Vehicle_No Slot_ID Entry_Time Exit_Time Duration
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HAS →	STORED IN →	
PARKING_SLOT PK Slot_ID Slot_Number Slot_Status	USED IN ↑	

Relationships: VEHICLE → PARKING_RECORD (1:N), PARKING_SLOT → PARKING_RECORD (1:N),
PARKING_RECORD → PARKING_HISTORY (1:1 for a completed record).